Egyptian Startup Intelligence System

Project documentation | DEPI R5 - Machine Learning Track
Planning version: 18 October 2026. Five-week implementation plan; handover on 22 November 2026.

1. Project Planning & Management

1.1 Project proposal

The project helps startup founders in Egypt understand their cash position, test business decisions, and find market information that applies to their situation. A founder enters a startup profile and asks a question. The system retrieves relevant evidence, calculates a scenario, or calls an evaluated prediction model when the available data supports that use.

For example, a founder can ask what happens to cash runway if selected expenses fall by 20%, then review the assumptions and related Egyptian market evidence. The numerical result comes from the calculation or model; the language model explains it and cites the retrieved material.

Problem and objectives

Useful information is spread across company databases, economic publications, funding reports, and a founder's own records. Local historical startup data is particularly limited. This makes it difficult to produce defensible predictions or compare decisions without mixing facts, assumptions, and estimates.

The team will build a working prototype that can: (1) answer a defined set of market questions with sources and dates; (2) compare 12-month cash scenarios; (3) complete a reproducible ML experiment on a clearly defined outcome; and (4) explain results through a simple dashboard and chat interface.

Scope for the five-week delivery

Included	Conditional or deferred
Startup profile, input validation, a curated RAG knowledge base, cash-flow simulation, dashboard, and integration tests.	A live ML prediction is enabled only if its inputs, labels, and evaluation are suitable. The evaluated research experiment remains a required deliverable.
Baseline, expense-change, cash-collection-change, and funding-delay scenarios. Source dates, assumptions, and model limitations remain visible.	Multi-year survival curves, reliable Egyptian failure probabilities, revenue-growth forecasting, and causal recommendations require stronger data.
Open or authorized data; a local or controlled demonstration; documented setup and evaluation.	Paid-data dependencies, bank integrations, automated investment decisions, and a production service are outside the initial delivery.

This document describes planned work and proposed acceptance targets. It does not report completed training, user interviews, or measured product performance.

1.2 Team responsibilities & resources

Each member owns a main work area and a concrete deliverable. Ownership keeps decisions moving; it does not prevent other members from contributing ideas, code, reviews, or support.

Team member	Main responsibility	Distinct focus and deliverables
Abdelrahman Ameen Team leader	System architecture and generative AI: RAG, orchestration, prompts, and response generation.	Own interfaces and scope decisions; maintain the RAG evaluation set; review integration; coordinate progress and the final demonstration.
Tarek Mohamed	Machine learning, data analysis, and database work.	Own data acquisition, cleaning, deduplication, the data dictionary, SQL schema, and source lineage. Prepare the first baseline with Youssef.
Youssef Taha	Machine learning and data analysis.	Own model comparisons, validation splits, experiment tracking, and ensemble evaluation. Produce the evaluation report and model card.
Karim Ahmed	Machine learning and data analysis.	Own feature engineering, explainability, and the financial simulation logic. Check formulas, economic assumptions, and scenario edge cases.
Abdelrahman Mohamed	User documentation and manual testing.	Write the short user guide, run agreed user-flow checks, record issues and feedback, and maintain the demo checklist. Support other members as needed.

Backend, dashboard, and integration work are shared by all five members, with Ameen coordinating the architecture and interfaces. Tarek, Youssef, and Karim share the ML and data-analysis workload. Abdelrahman Mohamed maintains user documentation and manual test records.

How the team will work

Use one task board with an owner, expected output, due date, and acceptance check for each task. Hold two short progress check-ins and one working demonstration each week. Record blockers early and reassign work when needed. All members participate in research, testing, documentation, and final presentation preparation.

Work on the relevant feature branch, open a pull request into develop, and request a peer review. Merge a tested integrated version into main at each agreed release. The existing branches are work areas; they do not all need to be active at once.

Resources and working assumptions

Data and tools: public datasets, authorized company samples, a small SQL store, a document index, and Python tools for ML and simulation. Select the backend, dashboard, and retrieval tools in Week 1. Keep raw and processed data separate.

Compute and access: use team laptops or available notebooks and benchmark before adding infrastructure. Set an LLM spending cap in Week 1. GitHub holds code, documentation, and permitted samples; credentials and private raw records remain outside the public repository.

1.3 Five-week implementation plan

Work begins on 18 October 2026. The five working weeks end on 21 November; final handover is planned for 22 November, five weeks after the start. These are the team's internal delivery dates.

Week / dates	Main work	End-of-week deliverable
Week 1 18-24 Oct	Agree supported questions and data fields. Audit datasets and licenses. Define the API contract, database schema, and RAG sources.	Ameen: architecture and question list. Tarek: EDA and data dictionary. All: agreed target shortlist and task board.
Week 2 25-31 Oct	Build retrieval, the cash-flow baseline, and the first ML experiment. Connect a basic form to working backend functions.	Ameen: sourced RAG answers. Karim: checked calculations. Youssef: baseline evaluation. Team: first integrated flow.
Week 3 1-7 Nov	Compare three model families on the same task. Evaluate an ensemble if justified. Add scenario controls and explanations.	Model comparison and deployment decision; baseline-versus-scenario chart; tested RAG-to-simulation flow. Tarek verifies data lineage.
Week 4 8-14 Nov	Integrate accepted components. Test routing, source citations, invalid inputs, privacy, and response times. Run usability sessions.	Team: feature-complete prototype and resolved high-priority defects. Abdelrahman Mohamed: manual test log, feedback, and draft user guide.
Week 5 15-21 Nov	Fix defects, rerun the frozen evaluation, complete the report and user guide, and rehearse the presentation.	Release candidate, reproducible experiments, final documentation, presentation, and a backup recorded demo.

Workstream schedule

Workstream	W1	W2	W3	W4	W5
Planning and requirements	Define	Refine	Review	Freeze	Report
Data and ML	Audit	Baseline	Compare	Validate	Report
RAG and orchestration	Sources	Build	Connect	Evaluate	Fix
Simulation	Rules	Build	Scenarios	Test	Fix
Backend and dashboard	Contract	Skeleton	Connect	Test	Release

Decisions that control the schedule

24 October: shortlist a usable outcome and identify missing data.
31 October: freeze the training target, feature definitions, and evaluation design. If the evidence cannot support a proposed question, narrow that question now.
7 November: decide whether the ML output is suitable for the user-facing demo or should remain an explicitly limited research result.
14 November: stop adding features; reserve the final week for quality and delivery.

1.4 Risk assessment & mitigation

The main risk is data suitability. The team will review this register each week and update the response when evidence changes. H/M denote a provisional high/medium likelihood and impact, not measured probabilities.

Risk / owner	Likelihood / impact	Response and decision trigger
Limited local startup data **Tarek + Ameen**	H / H	Start with Egypt and MENA, then expand to compatible emerging-market and global data. Reduce the supported questions if necessary. Seek voluntary samples from companies known to the team. Decide on the initial scope by 31 Oct.
Missing dates, unclear outcomes, or leakage **Youssef**	H / H	Define the prediction date and outcome before training. Remove post-outcome fields and deduplicate companies. Without reliable event dates, do not claim time-to-event or fixed-horizon failure predictions.
Global data does not transfer to Egypt **Youssef + Tarek**	H / H	Record country, sector, stage, and year. Evaluate relevant subgroups where sample size permits. Increasing global coverage does not automatically validate local performance; restrict claims and seek local checks.
Data access or reuse restrictions **Tarek**	M / H	Document license and provenance. Prefer clearly open sources. Treat unresolved Crunchbase mirrors and YC as provisional research inputs, and keep paid or application-gated sources off the critical path.
Small sample or weak ML performance **Youssef**	H / H	Start with a simple baseline and controlled validation. Use fewer defensible features. Keep the best evaluated single model if an ensemble adds no benefit. Disable unsupported live predictions and report the experiment honestly.
Unsupported RAG answers or stale evidence **Ameen**	M / H	Retain publisher, URL, document date, and retrieved passage. Test citations and abstention. Refresh selected sources; show the evidence date and state when the corpus cannot answer a question.
Incorrect scenario assumptions **Karim**	M / H	Use explicit cash receipts, payments, timing, and currency. Check formulas against hand calculations. Show edited assumptions and distinguish cash-flow effects from any claim about failure probability.
Integration delays or uneven workload **Ameen + team**	M / H	Agree interfaces in Week 1 and connect a working flow in Week 2. Review ownership weekly. Reduce optional scenarios before cutting validation or the final integration window.
Private records, API cost, or service outage **Ameen + team**	M / H	Collect company data only with agreed permission and use. Remove identifiers, restrict access, and exclude raw records from GitHub. Limit model calls, log spending, and retain a local demo dataset and deterministic simulation fallback.

How local company samples will help

Company contacts can supply authorized monthly figures and test whether the workflow is useful. A few friendly companies are valuable case studies, but they do not establish an unbiased Egyptian training dataset. If outcomes are missing, use the samples for scenario validation and usability testing rather than inventing failure labels.

1.5 Success measures & evaluation

The targets below are proposed acceptance criteria for the prototype. They are not results already achieved. Freeze the test questions, manual scenarios, and model-validation protocol by the end of Week 2.

Area	Target / evidence	Owner
Data readiness	Every selected dataset has a source, usage status, row definition, target definition, missingness report, and record of preprocessing. Every model feature has an availability check.	Tarek
Retrieval	On 20 answerable held-out questions, retrieve an expert-marked relevant passage in the top five for at least 16 questions. Keep separate development questions for tuning.	Ameen
Answer grounding	At least 90% of checked factual claims are supported by their cited passages. All cited sources resolve to stored metadata; unsupported statements are corrected.	Ameen + reviewer
Insufficient evidence	On 10 held-out questions the corpus cannot answer, at least 9 responses clearly state the limitation instead of inventing an answer.	Ameen
Simulation correctness	All 10 hand-checked baseline and edge-case scenarios match expected monthly cash balances within 0.01 reporting-currency units, using the same rounding convention.	Karim
ML experiment	Document a simple reference baseline, three model families, and the ensemble decision on the same target and evaluation protocol. Report uncertainty and class counts; accuracy alone is insufficient.	Youssef
Interactive performance	Proposed p95 targets: simulation within 2 seconds and RAG within 20 seconds. Measure at least 30 requests on declared hardware/network conditions with up to 5 concurrent users.	Whole team
Usability and delivery	At least 4 of 5 testers complete the profile, scenario comparison, and source-checking tasks without developer intervention. All must-have acceptance checks pass before handover.	Whole team

Model evaluation rules

Compare Logistic Regression, Random Forest, and a gradient-boosting model after a simple reference baseline. Split by company when records repeat, and use temporal validation when the data supports it. Fit preprocessing and any resampling within training folds. Use out-of-fold predictions for stacking; never train the stacker on in-sample model predictions.

For an imbalanced failure or bankruptcy target, report average precision, precision, recall, F1, and a confusion matrix at a stated threshold. Check calibration before presenting probabilities. Select models using validation data and evaluate the selected approach on untouched data where feasible; small samples may require a clearly described nested or repeated validation design.

An ensemble is optional. If results are unstable or fail to improve on a relevant baseline, retain the analysis as a research finding. Numerical ML thresholds will be agreed after the baseline and before final evaluation; they will not be chosen retrospectively to make a result pass.

2. Literature Review

2.1 Existing products and approaches

This review uses official product descriptions and research publications. Product capabilities are vendor-reported; the team has not independently benchmarked these services. Limitations below describe their fit to this project, rather than a claim that a product lacks every unlisted feature.

Dealroom

Dealroom provides company, investor, funding, and ecosystem intelligence, with reported and estimated business indicators and predictive funding signals. Its breadth and linked company histories are useful for market research. Full search, exports, and API access depend on the access plan, so the project cannot assume an unrestricted training dataset. The lesson is to retain source lineage and distinguish reported values from estimates. [1]

Crunchbase

Crunchbase offers private-company profiles, funding histories, investor information, growth signals, and monitoring. Its Pro page also distinguishes Business-level predictions from Pro features. It is relevant to company discovery and benchmarking, but access to a product is different from permission to redistribute or train on its data. The project will make the outcome definition and training population visible rather than treating an external score as ground truth. [2]

LivePlan

LivePlan combines business planning with cash-flow forecasts, financial scenarios, and comparisons with actual results. It also describes conversational assistance for building forecasts. This is the closest product reference for the scenario workflow. Its usefulness depends on the business inputs and assumptions supplied; the reviewed feature page does not establish the Egyptian startup failure model proposed here. The project can adopt clear baseline comparisons and explicit cash timing. [3]

2.2 Comparison with the proposed system

Approach	Useful strength	Project emphasis
Dealroom [1]	Broad ecosystem and company intelligence.	A small, traceable evidence base focused on selected Egyptian startup questions.
Crunchbase [2]	Company histories, discovery, and monitoring.	Transparent data selection, clearly defined targets, and reproducible model evaluation.
LivePlan [3]	Cash forecasts and side-by-side what-if scenarios.	A limited scenario engine connected to dated market evidence and an evaluated ML component where appropriate.

These products overlap with parts of our design. The proposed contribution is a focused educational implementation for the chosen users and available data, not a claim that no integrated business-planning tools exist.

2.3 Research foundations & contribution

Startup outcome prediction

Potanin et al. study success milestones for Series B and Series C startups using Crunchbase data and a simulated venture-capital investment process. Their focus on IPO, unicorn, and acquisition outcomes shows why success must be defined precisely. That investor setting differs from predicting closure or cash pressure for an early Egyptian startup. We will use explicit outcome definitions and time-aware evaluation rather than transfer the paper's reported performance. [4]

Maarouf et al. combine startup descriptions and structured information in a model evaluated on 20,172 Crunchbase profiles. The work suggests that business descriptions can add information, but requires suitable data access and its own outcome definition. Text-based prediction is a possible later extension; the first five-week experiment will prioritize a manageable tabular pipeline. [5]

Retrieval-augmented generation

Lewis et al. combine a retriever and an external document index with a language-generation model. This provides a basis for retrieving evidence before answering a market question. The project will adapt the approach to a curated corpus, preserve source dates, and test whether answers are supported. Retrieval does not remove the need to evaluate missing, stale, or irrelevant evidence. [6]

Explainability and RAG evaluation

Lundberg and Lee introduce SHAP as a method for assigning feature contributions to individual predictions. We will use model explanations to show influential inputs where feasible. Contributions describe the model, and will not be presented as proof that changing a feature causes a business outcome. [7]

RAGAs evaluates different aspects of retrieval-augmented answers. It is a useful reference for separating retrieval quality from answer faithfulness. Our evaluation will include human-checked passages and claims, with automated scores as supporting evidence. [8]

2.4 Proposed contribution and lessons applied

The prototype connects three activities that a founder can inspect: looking up dated evidence, calculating a cash scenario, and viewing a model result only when the model fits the available inputs and intended population. Egyptian economic information supports both retrieval and scenario assumptions. An orchestrator decides which components to call and the order of those calls.

The contribution will be judged by the quality of the working workflow and its evaluation. A global experiment will be labeled as global; an old cohort will retain its observation period; a scenario will display its assumptions. The team will not claim reliable local failure probabilities before local validation exists.

2.5 Improvements after the first delivery

Collect authorized local histories, expand verified outcome labels, evaluate additional sectors, and study growth or survival models when repeated observations and dates become available. Document reviewer feedback in the issue tracker and prioritize changes by evidence, user value, and effort.

3. Requirements Gathering

3.1 Stakeholders and needs

Stakeholder	Needs and expectations
Startup founders and operators	Understand cash runway, compare spending and funding assumptions, and find relevant market evidence without reading many reports.
Mentors and incubator staff	Review assumptions, inspect source material, and discuss results with founders. They need traceability rather than an unexplained score.
Companies sharing data	Understand the agreed purpose, access boundaries, retention, and whether their records will appear in a demonstration or publication.
Development team	Clear interfaces, reproducible datasets, testable requirements, and visible decisions about model limitations.
Technical instructor and evaluators	A defensible method, working demonstration, source documentation, evaluation results, and evidence of individual contributions.

How requirements will be checked

The initial requirements come from the project brief, team discussions, and the data audit. During Week 1, seek short conversations with three to five accessible founders, mentors, or suitable proxy users. Ask which decisions matter, which financial fields they can supply, and how they check evidence. These conversations are planned, not completed research.

Record each proposed change with its source and effect on scope. Ameen maintains the accepted requirement list; the whole team reviews it. Week 4 usability sessions will check the implemented workflow against the original needs.

3.2 User stories

ID	User story	Related requirements
US-01	As a founder, I want to enter dated financial figures and receive clear validation messages so that calculations use the right inputs.	FR-01, FR-02
US-02	As a founder, I want an answer with sources and dates so that I can check whether the market evidence applies to my company.	FR-03 to FR-05
US-03	As an operator, I want to compare a baseline with an expense, collection, or funding change so that I can see the cash effect.	FR-06, FR-07
US-04	As a mentor, I want to see a prediction's target, inputs, explanation, and limits before discussing it with a founder.	FR-09, FR-10
US-05	As a team member, I want to update sources and reproduce experiments so that the system can be reviewed and maintained.	FR-08, FR-11, FR-12

3.3 Functional requirements

Must: required for the delivery. **Conditional:** implemented for the user-facing model only if the Week 3 data and evaluation decision permits it. A documented ML experiment is still mandatory.

ID / priority	Requirement	Acceptance evidence
FR-01 Must	Accept a startup profile, question, reporting currency, and observation date, plus the fields required for the selected scenario.	A valid example is accepted and its units and date remain visible in the result.
FR-02 Must	Validate required fields, ranges, dates, currencies, and consistency. Identify missing information without fabricating values.	Invalid and incomplete examples receive specific field-level messages.
FR-03 Must	Route supported questions to RAG, simulation, or a combined flow. Add prediction routes only for an accepted model.	A labeled route test set covers market-only, scenario-only, combined, and unsupported requests.
FR-04 Must	Retrieve relevant passages from the approved corpus using topic, country, and date metadata where available.	Retrieval is checked against the marked passages in the held-out question set.
FR-05 Must	Generate evidence-based answers with source links, document dates, and a clear response when evidence is insufficient.	Citation and abstention targets in Section 1.5 are met.
FR-06 Must	Calculate monthly cash for a 12-month baseline and supported expense, cash-collection, and funding-delay scenarios.	Hand-checked cases pass; calculations use the entered timing and amounts.
FR-07 Must	Display cash paths, cash runway, the first minimum-buffer breach, funding gap, and changed assumptions.	Dashboard values match engine outputs and support a side-by-side comparison.
FR-08 Must	Support reproducible dataset preparation, a baseline, model comparisons, and an ensemble decision.	Versioned experiments and an evaluation report can be rerun from the documented setup.
FR-09 Conditional	Return predictions only for supported feature sets and populations; identify the target and any justified forecast horizon.	The response includes model version and limitations; incompatible inputs are rejected.
FR-10 Conditional	Explain the accepted model's output using appropriate feature contributions and state their limits.	Explanations refer to the actual inputs and model; causal claims are not inferred from SHAP.
FR-11 Must	Allow the project team to ingest and refresh approved documents with publisher, URL, date, and usage metadata.	A new source can be added and traced from a retrieved passage to its original document.
FR-12 Must	Record reproducibility metadata for a run: input/scenario version, engine/model version, corpus version, and source references.	A permitted sample result can be reproduced; private raw values are excluded from public logs.

3.4 Non-functional requirements

ID / quality area	Requirement	Verification
NFR-01 Performance	Aim for p95 simulation latency of 2 seconds and RAG latency of 20 seconds under the declared prototype workload.	Benchmark at least 30 requests, up to 5 concurrent users; record hardware, network, model, and warm/cold behavior.
NFR-02 Reliability	Handle missing fields, invalid values, empty retrieval, and external-service failures without unhandled crashes.	Run at least 10 invalid/boundary cases and simulate a retrieval or model-service failure.
NFR-03 Privacy and security	Keep secrets and private raw records out of GitHub. Restrict access to volunteered company data and minimize information sent to external models.	Review configuration, repository contents, access settings, and sample logs before any shared demo.
NFR-04 Source handling	Treat retrieved documents as evidence, not instructions. Validate model tool arguments and keep the permitted tool set narrow.	Test malicious instructions embedded in documents and invalid tool-call parameters.
NFR-05 Usability	Use clear labels, currencies, dates, and assumptions. Distinguish retrieved facts, calculated scenarios, and model outputs.	At least 4 of 5 testers complete the defined tasks without developer intervention.
NFR-06 Reproducibility	Record dataset versions/checksums, preprocessing, seeds, dependencies, model configuration, and evaluation splits.	Another member follows the setup and reproduces the main experiment and a sample scenario.
NFR-07 Maintainability	Keep retrieval, simulation, ML, routing, storage, and interface code in separate modules with documented contracts.	Peer review and integration tests verify that one component can change without silently changing the others' outputs.
NFR-08 Evidence quality	Use metrics appropriate to the task and report sample limits, subgroup coverage, and uncertainty. Do not turn a scenario assumption into an asserted forecast.	Review model cards, RAG test results, scenario descriptions, and final presentation claims.

Verification approach

Karim maintains reference calculations; Youssef owns model evaluation; Ameen maintains RAG and routing tests; Tarek verifies data versions. Abdelrahman Mohamed runs agreed manual user-flow checks and records feedback. The whole team shares software integration and automated testing. A second member reviews each area before acceptance.

Acceptance targets may be refined after the Week 2 benchmark, with the reason recorded. They must be fixed before the final test run. Any failed target appears in the final limitations and issue list rather than being silently removed.

3.5 Data requirements & collection strategy

Prefer suitable Egyptian and regional data. When a regional equivalent is missing or too weak for a particular target, evaluate European and other international data with equal acquisition priority. This does not imply equal training weights or permission to combine incompatible labels.

Data group	Available evidence / purpose	Required checks
Startup cohort	Spain technology ventures: 175 firms, with 150 observed survival labels. A candidate for a limited historical startup experiment. [9]	Check variable timing, missing targets, survey measures, and sample size. Exclude 2014 outcome fields from earlier predictors.
Financial classification	Taiwan: 6,819 records. Poland: 43,405 financial statements across five files. Separate accounting-bankruptcy experiments. [10] [11]	Audit units, class imbalance, company overlap, and prediction horizons. Do not equate bankruptcy with every type of startup failure.
Company and event discovery	Crunchbase snapshots, IdeaProof events, and YC profiles help investigate coverage and candidate labels. [13] [14] [15]	Confirm usage rights and event definitions. Duplicates, Inactive status, and distress reports cannot automatically become shutdown labels.
Economic context	World Bank country-year indicators are already extracted. CBE and other official publications are candidate local sources. [12]	Match country and observation time; record release dates. Annual FX is not a spot rate, and lending rates are not policy rates.
RAG documents	Selected economic publications, sector reports, and other authorized startup material.	Retain title, publisher, URL, publication date or date-unknown flag, retrieval date, geography, sector, and usage status.
Voluntary company records	Monthly figures from willing contacts can support realistic scenario cases and usability testing.	Agree purpose and access, remove identifying details, retain only necessary fields, and record missing or self-reported values honestly.

Local-data fallback

If local labels remain insufficient, narrow the prediction questions, expand to compatible global data, and keep Egyptian evidence in the RAG and scenario context. Request authorized company samples where practical, but do not make the delivery depend on receiving them. A handful of companies cannot establish local predictive accuracy.

Data preparation and storage

Keep raw and processed versions separate. Use stable company identifiers where possible; document duplicate decisions and preserve original units. A small SQL schema can hold company metadata, dated observations, source records, and scenario runs. Store document text and retrieval metadata in the document index. Fit learned preprocessing only on the training split.

Dataset counts refer to our 8 October 2026 audit. Availability, licensing, and suitable targets will be rechecked before the Week 2 training freeze.

3.6 Inputs, outputs & main use cases

Inputs	Outputs
Profile: country, sector, business model, currency, observation date, and question.	Selected route, a clear answer, sources and dates, and any missing information.
Scenario: opening cash, minimum cash buffer, monthly receipts and payments, funding amount/date, and the assumptions being changed.	12-month cash path, baseline comparison, minimum-buffer breach, funding gap, and assumptions.
Optional extensions: cash-collection timing, category costs, one-off payments, and FX exposure with an explicit rate assumption.	Additional scenario comparisons only where the required inputs and calculation rules are available.
ML: the exact approved training features, definitions, units, and observation timing.	A supported model outcome, version, explanation, and applicable limitations. A horizon is shown only if supported by the target.

UC-01: Ask a market question

The founder asks a supported question and supplies relevant country or sector context. The orchestrator calls retrieval; the system returns an answer linked to evidence and its dates. If passages do not support an answer, it explains the gap and identifies what information is needed. Related requirements: FR-03 to FR-05.

UC-02: Compare a cash scenario

The founder supplies validated cash inputs, chooses a change, and sets its timing. The engine calculates the baseline and alternative month by month. The dashboard shows the cash paths and buffer breach. Missing cash or timing information is requested before calculation. Related requirements: FR-01, FR-02, FR-06, FR-07.

Calculation rule: closing cash = opening cash + cash receipts + funding inflows - cash payments. Closing cash becomes the next month's opening cash. Payments include any relevant debt service, tax, capital spending, and one-off costs; avoid counting the same item twice. Funding raised historically is not current cash, and booked revenue is not necessarily cash collected.

Report the first month with negative closing cash as the projected cash-shortfall point, and flag the minimum-buffer breach separately. If cash stays non-negative, report runway beyond the 12-month horizon rather than extrapolating an unlimited runway.

UC-03: Request an assessment with an explanation

When an accepted model exists, the system checks whether the supplied profile matches its feature definitions and supported population. It returns the model output and explanation, then adds relevant retrieved evidence if requested. If the model is unsuitable, it states the limitation and offers supported retrieval or scenario analysis. Related requirements: FR-03, FR-09, FR-10.

3.7 Assumptions and boundaries

The first version serves a small prototype workload and a limited set of agreed questions. English is the baseline interface and evaluation language; Arabic support can be added if tested within the schedule. Company samples and additional sources are optional. Survival curves and growth forecasts require adequate dates and repeated observations. Synthetic examples may test software behavior, but cannot demonstrate real predictive performance.

The architecture remains: user input, orchestrator, RAG and/or prediction/simulation, explanation where relevant, response generation, and dashboard. Egyptian economic context is a supporting data layer. Scenario changes are assumptions, and feature attributions do not establish causal effects.

References

Product pages and online documentation were reviewed on 18 October 2026. Dataset observations refer to the local audit dated 8 October 2026. Links below identify the sources used; product features and access arrangements may change.

[1] Dealroom (2026). Dealroom FAQ: coverage, signals, data access, and plans.
https://dealroom.co/faq/

[2] Crunchbase (2026). Crunchbase Pro: private company data and product features.
https://about.crunchbase.com/products/crunchbase-pro

[3] LivePlan (2026). Financial Forecasts & Scenarios.
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[4] Potanin, M., Chertok, A., Zorin, K., & Shtabtsovsky, C. (2023). Startup success prediction and VC portfolio simulation using CrunchBase data. Preprint.
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[5] Maarouf, A., Feuerriegel, S., & Prollochs, N. (2024). A Fused Large Language Model for Predicting Startup Success. European Journal of Operational Research.
https://arxiv.org/abs/2409.03668

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[9] Garcia-Cabrera, A. M., & Garcia-Soto, M. G. (2024). Technology-based entrepreneurs database. Zenodo.
https://doi.org/10.5281/zenodo.10607031

[10] Tomczak, S. (2016). Polish Companies Bankruptcy. UCI Machine Learning Repository.
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[11] Taiwanese Bankruptcy Prediction (2020). UCI Machine Learning Repository.
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https://datatopics.worldbank.org/world-development-indicators/

[13] Kaggle / justinas. Startup Investments: historical Crunchbase data mirror.
https://www.kaggle.com/datasets/justinas/startup-investments

[14] IdeaProof. Startup datasets and failure database.
https://ideaproof.io/datasets

[15] YC OSS. Unofficial Y Combinator companies API.
https://github.com/yc-oss/api

Project repository: github.com/Abdelrhman-Ameen/egyptian-startup-intelligence-system